Tu2C-4

A 56 GHz Trilayer AlN/ScAlN/AlN Periodically Poled FBAR

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Outline

- Introduction
- Structural Design and Analysis
- Film Growth and Device Fabrication
- Poling and Resonator Measurements
- Conclusion

Acoustic Filters for mm-wave frontends

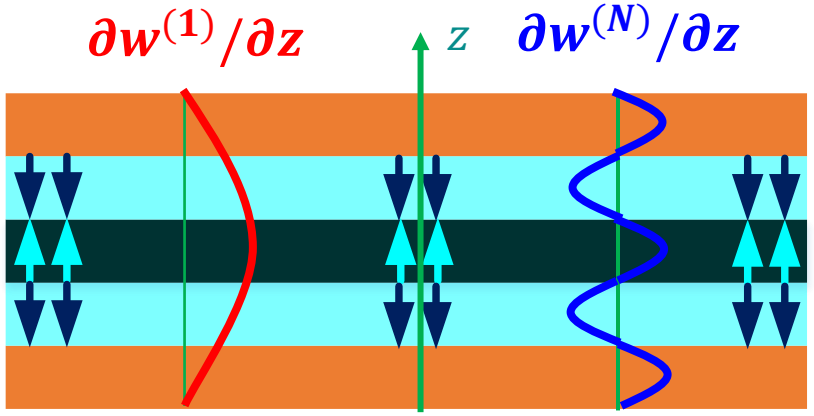
- New bands at higher frequencies require advanced filters
- Acoustic filters: technology in cellphone frontend filters for sub-6 bands
- Challenges for scaling conventional FBARs to mm-waves
  - Reduced film thickness and film quality
  - Increased loss and reduced coupling factor
- Innovations in device design and material growth enable acoustic filters at mm-wave frequencies
  - Periodically poled FBAR
  - MBE grown AlN and ScAlN

5G FR2 Bands

Band	Frequency (GHz)	Bandwidth (MHz)
n257	26.50 - 29.50	3000
n258	24.25 - 27.50	3250
n259	39.50 - 43.50	4000
n260	37.00 - 40.00	3000
n261	27.50 - 28.35	850
n262	47.20 - 48.20	1000
n263	57.00 - 71.00	4000

Periodically Poled FBAR

- Coupling factor is a function of material constants and mode shapes
- Mode shapes and resonance frequencies: eigenfrequency analysis
- Periodically Poled FBAR: efficiently operates in mode N
- Maintains overall device thickness
- Promising approach for preserving k_t^2 and Q at mm-waves



Mode shapes in an FBAR

Displacement $w^{(n)}$

Strain $w^{(n)}_{,z} \equiv \frac{\partial w^{(n)}}{\partial z}$

$D = 0$: $w^{(n)}$ and ω_n satisfy

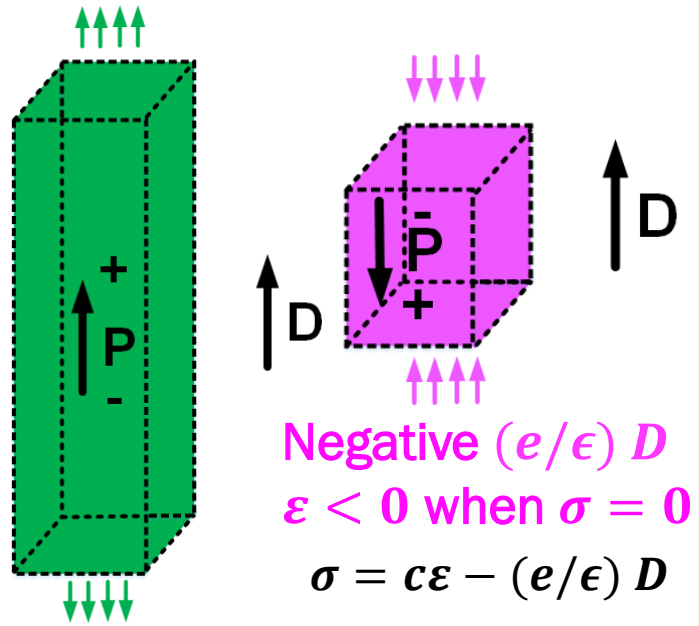
$$\rho\omega_n^2 w^{(n)} = \left(c w^{(n)}_{,z}\right)_{,z} \equiv \frac{\partial\left(\frac{c\partial w^{(n)}}{\partial z}\right)}{\partial z}$$

"Eigenfrequency analysis"

Polarization Switchable ScAlN

- ScAlN: higher $|e|$ than AlN
- Ferroelectric ScAlN can be poled to either expand or contract (under zero stress) given the same electric flux density $\boldsymbol{D}$
- Mode shapes are normalized and orthogonal:
  - $\langle w^{(n)} | \rho w^{(m)} \rangle = \delta_{mn}$
  - $\left\langle \frac{\partial w^{(n)}}{\partial z} \middle| c \frac{\partial w^{(m)}}{\partial z} \right\rangle = \omega_n^2 \delta_{mn}$
- Goal: periodically polarize multiple layers, optimized for one higher order mode

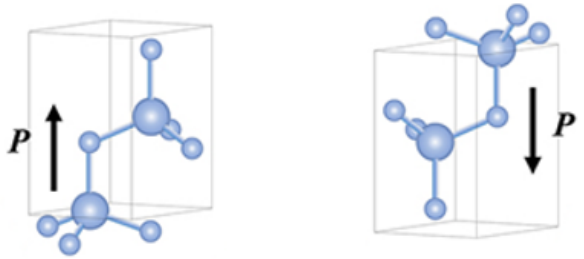
$$k_t^2 = \frac{\pi^2}{8} \frac{\omega_p^2 - \omega_s^2}{\omega_p^2}$$



Negative $(e/\epsilon)\ D$
$\varepsilon < 0$ when $\sigma = 0$

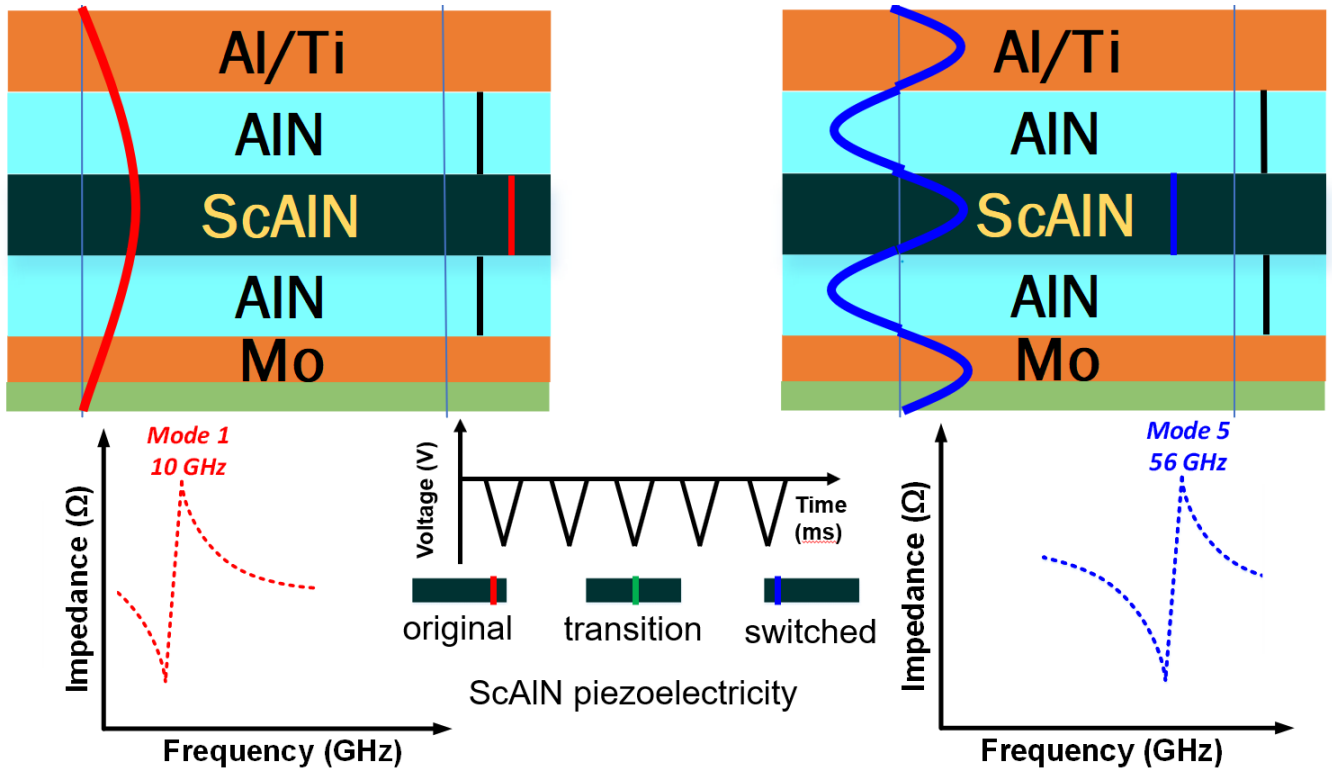
$\sigma = c\varepsilon - (e/\epsilon)\ D$

Positive $(e/\epsilon)\ D$
$\varepsilon > 0$ when $\sigma = 0$



Trilayer AlN/ScAlN/AlN FBAR

- MBE grown single crystal wurtzite phase nitride layers in one step
- Periodically poled nitride layers via polarization switching
- Layer thicknesses optimized for the fifth thickness extensional mode



"Frequency response": modal analysis

Modal expansion $w = \sum_n b_n(t) w^{(n)}(z)$

Modal amplitude $b_n(t) = D \left\langle \frac{\partial w^{(n)}}{\partial z} \middle| e/\epsilon \right\rangle / (\omega_n^2 - \omega^2)$ where $D = D_0 \cos(\omega t)$

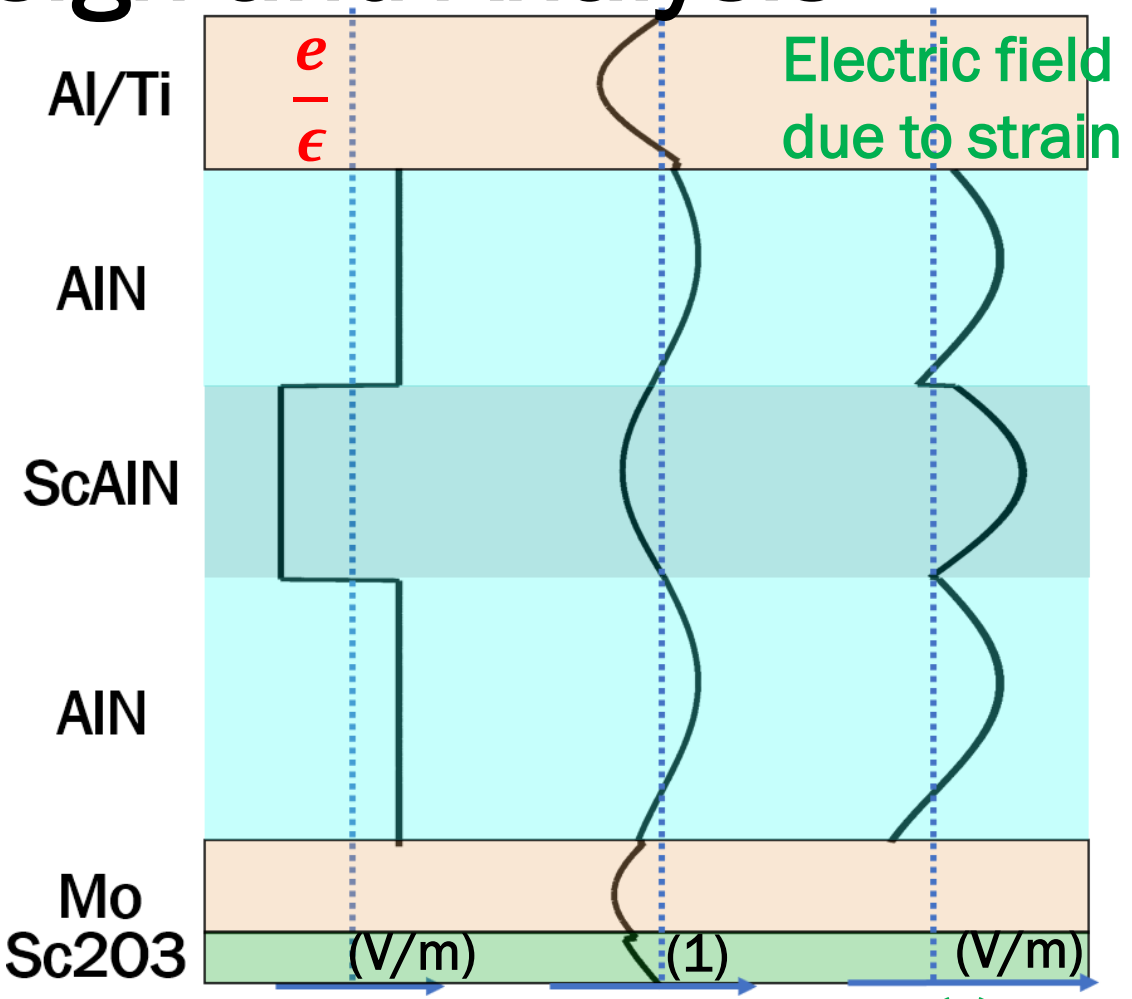
1) Correlation between modal strain and piezoelectric constants; 2) spacing in frequency

$$k_t^2 = \frac{\pi^2}{8} \frac{\omega_p^2 - \omega_s^2}{\omega_p^2}$$

Structural Design and Analysis

- Polarization switched ScAlN and as grown AlN work together efficiently only in the fifth mode
- f_p =58 GHz, k_t^2 =4.2% from COMSOL simulation

$$k_{t,n}^2 = \frac{\pi^2}{8}\frac{\omega_p^2-\omega_s^2}{\omega_p^2} = \frac{\pi^2}{8}\left(\frac{\left\langle w_{,z}^{(n)}\middle| e/\epsilon\right\rangle^2}{\int(1/\epsilon)dz\left\langle w_{,z}^{(n)}\middle| cw_{,z}^{(n)}\right\rangle}\right)$$



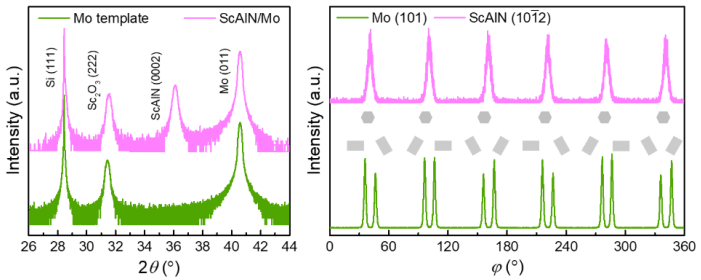
Electric field due to TE5 modal strain

Has the same sign (mostly)

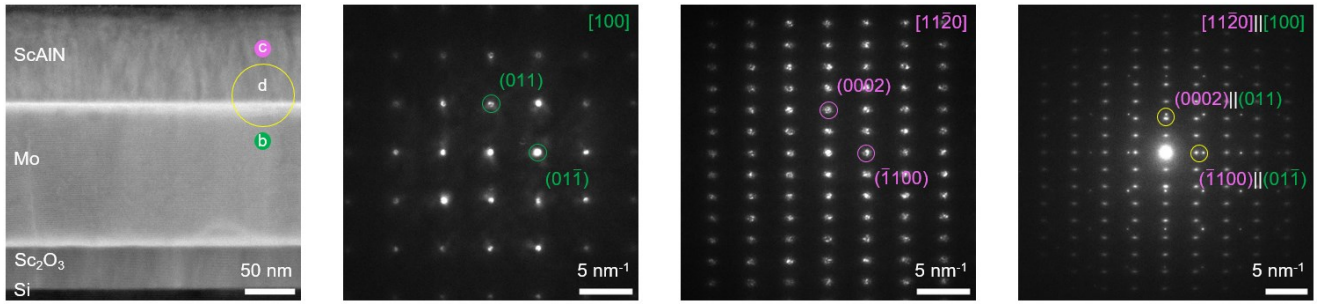
Piezoelectric constants $\frac{\partial w^{(5)}}{\partial z}$ $\left(\frac{\partial w^{(5)}}{\partial z}\right)\left(\frac{e}{\epsilon}\right)$

TE5 Modal strain

MBE Growth of Single Crystal Nitride

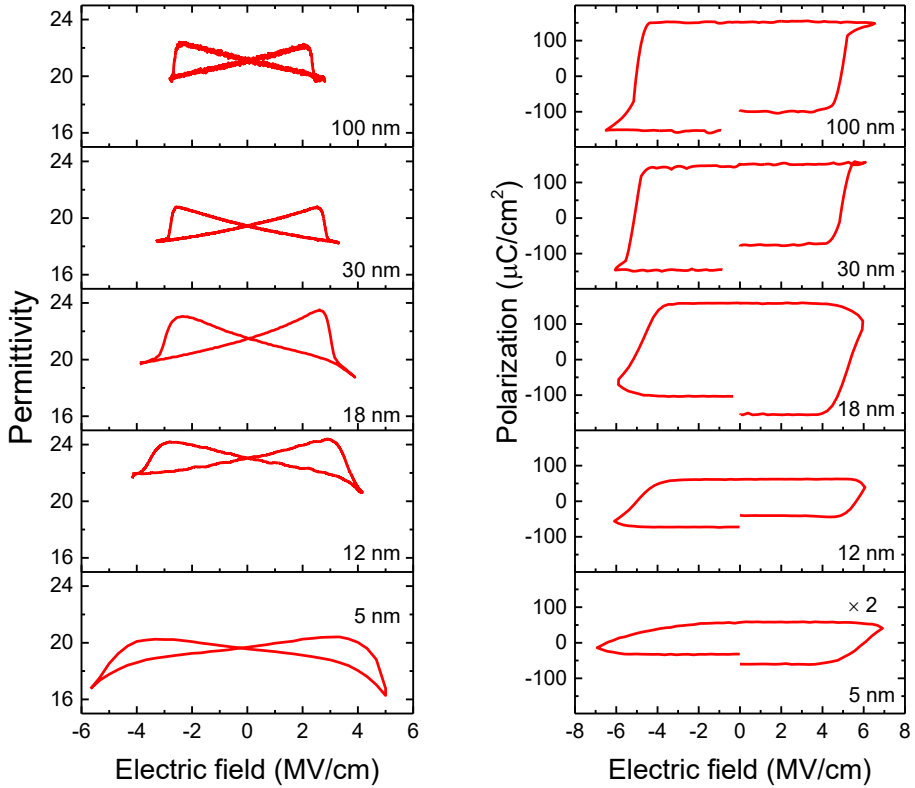


ScAlN can epitaxy grow on three types of Mo domains



Single-crystalline wurtzite phase ScAlN was achieved on polycrystalline Mo surface

ACS Applied Materials & Interfaces 15.14 (2023): 18022-18031.

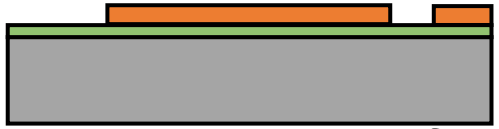


Ferroelectricity verified in single-crystalline ScAlN with high Sc contents down to 5 nm

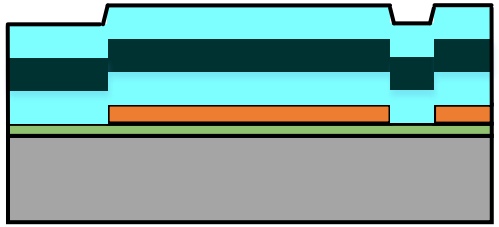
Appl. Phys. Lett. 122, 052101 (2023)

The trilayer AlN/ScAlN/AlN in this work is MBE grown using the optimized conditions developed in these works.

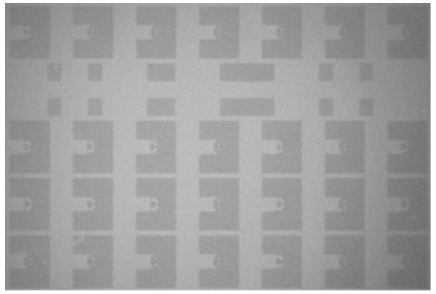
Film Growth and Device Fabrication



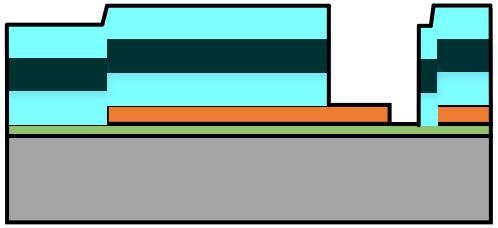
1. XeF_2 patterning of Mo bottom electrode



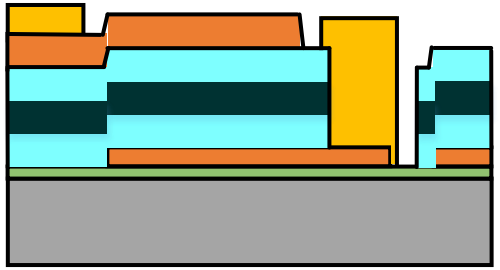
2. MBE growth of trilayer AlN/ScAlN/AlN



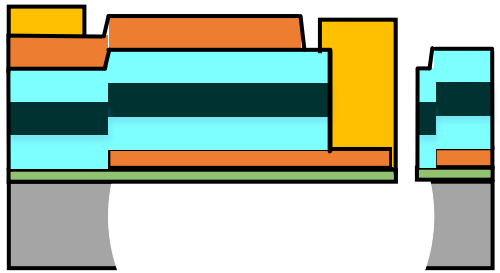
High quality nitride growth on Mo verified by SEM



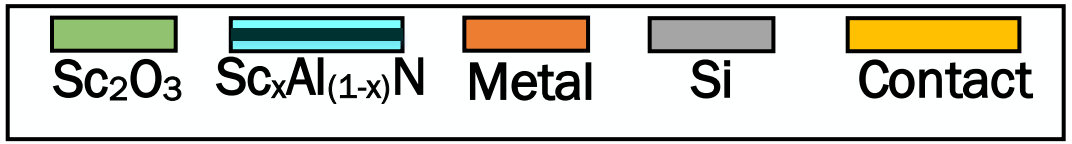
3. Ion Mill and wet etching of the via holes in the $Sc_xAl_{(1-x)}N$ layers



4,5. PVD top electrode and contact layers, patterned by liftoff

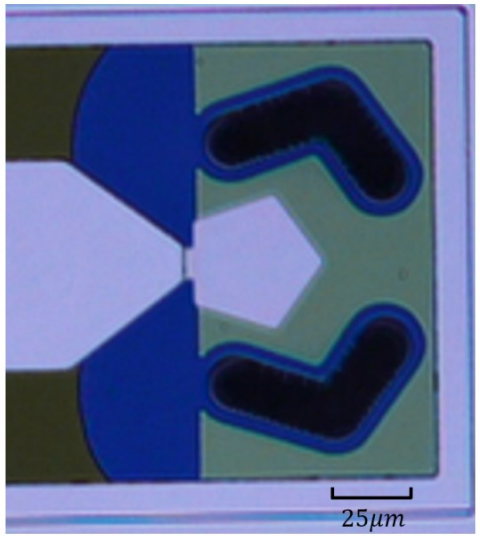


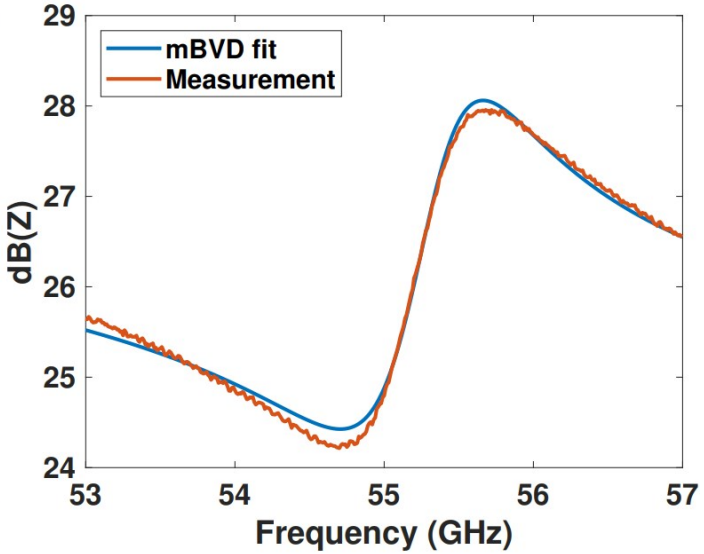
6,7. Ion Mill Sc_2O_3
XeF_2 isotropic Si etch

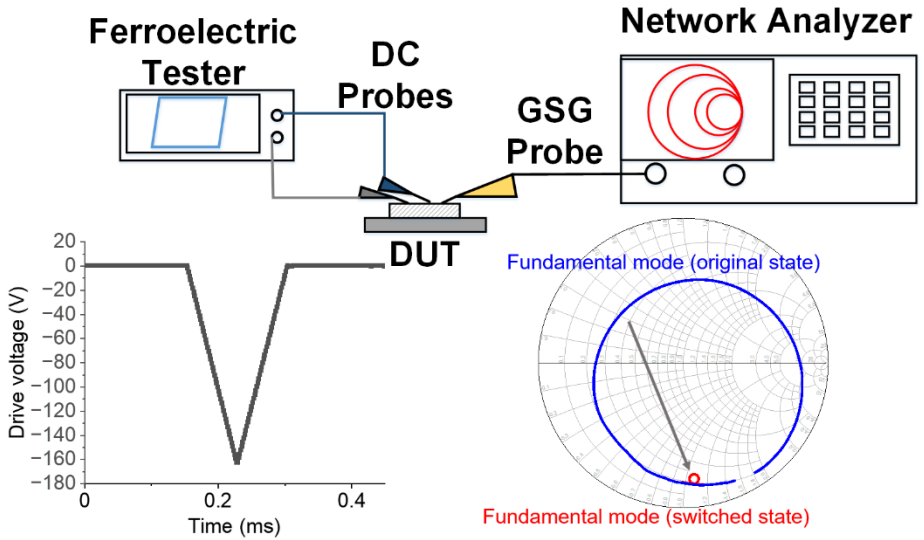


Poling and Resonator Measurements

- Poling progress monitored by frequency response
- Fundamental mode ↓ while fifth mode ↑
- Fully poled state: resonance at 56 GHz with $k_t^2 = 4.2\%$ and $Q_m = 60$
- Comparable FoM for FBAR (>40GHz)





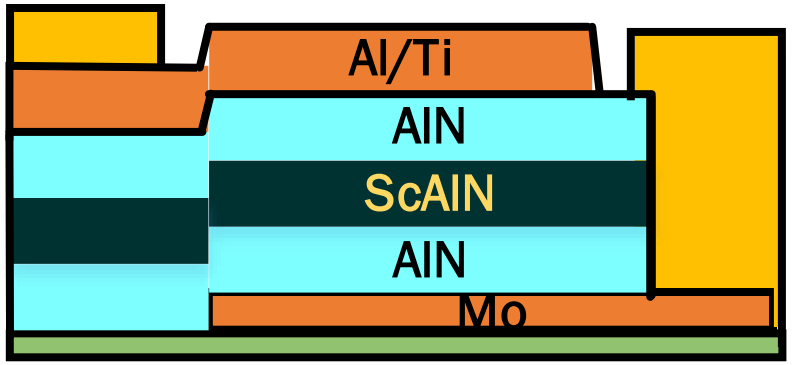


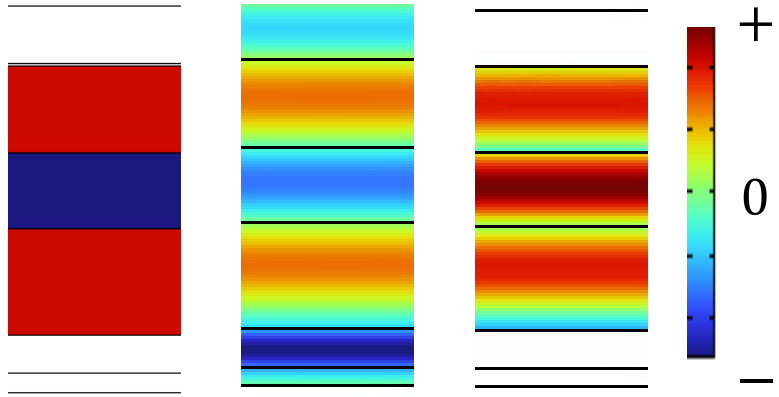
Comparison with other AlN/ScAlN based mm-wave FBARs

Work	f (GHz)	Q_m	k_t^2 (%)	$Q_m \cdot k_t^2$
This	***55.7***	***60***	***4.2***	***2.5***
[13]	55.4	30	3.8	1.1
[14]	55	90	2.2	2.0
[12]	33	110	1.7	1.9
[22]	31	61	5.5	3.4

Conclusion

- A new trilayer AlN/ScAlN/AlN periodically poled FBAR
- Successfully demonstrated poling and 56 GHz measurements for the fifth mode
- Future work on enhancing Q and k_t^2





Piezoelectric constants $\frac{e}{\epsilon}$

Modal strain $\frac{\partial w^{(5)}}{\partial z}$

Electric field due to strain $\left(\frac{\partial w^{(5)}}{\partial z}\right)\left(\frac{e}{\epsilon}\right)$